

VIX Regime Classification — Research Summary

Akshat Gupta | TAMS, University of North Texas | Advisor: Prof. Jianguo Liu | June 2026

Overview

This summary reflects the current state of the paper as of June 2026. The paper has been significantly reframed since the May 2026 summary. It began as a positive ML result — showing that selective prediction achieves 90–93% accuracy on VIX regime classification. Deeper analysis revealed the accuracy is entirely driven by VIX autocorrelation, not genuine forecasting skill. The paper is now a diagnostic decomposition that formally identifies and tests this 'persistence illusion' through a six-component protocol, replicated across three tasks: VIX regimes, S&P; 500 trend regimes, and yield curve inversion.

Key Changes Since May 2026 Summary

Pipeline: RandomizedSearchCV hyperparameter tuning removed. All results use a fixed-spec calibrated pipeline: RF (n_estimators=200, max_depth=10, min_samples_leaf=5) and HGB (max_iter=200, max_depth=5, lr=0.1), both with isotonic calibration via CalibratedClassifierCV (5-fold TimeSeriesSplit). XGBoost replaced by HistGradientBoosting (HGB) throughout.

Framing: Positive result framing removed. The headline accuracy (90–93%) is shown to be an artifact of VIX autocorrelation. A 3-feature logistic regression matches the 36-feature RF at 4 of 5 horizons. No RF-vs-persistence gap reaches statistical significance under moving-block bootstrap.

New sections added: (1) Regime transition analysis — 0% accuracy on calm→high spike days at every horizon, even with cost-weighting up to 20×. (2) S&P; 500 and yield curve cross-task replications. (3) Synthetic positive control confirming the diagnostic has power to detect real signal.

Main Results (VIX ≥ 20, τ = 0.25)

N	RF Baseline	RF Selective	RF Gain	RF Coverage	RF BalAcc	HGB Sel	HGB Gain	HGB Cov
5	83.4%	91.9%	+8.5 pp	76.3%	88.4%	92.1%	+8.9 pp	71.1%
10	79.5%	90.0%	+10.5 pp	60.1%	84.8%	92.2%	+13.6 pp	63.1%
15	76.2%	90.0%	+13.8 pp	51.3%	83.9%	92.2%	+21.3 pp	36.0%
20	76.8%	92.7%	+15.9 pp	31.7%	79.1%	94.8%	+22.2 pp	26.8%
25	73.4%	90.3%	+16.9 pp	26.9%	50.0%*	92.7%	+26.5 pp	19.2%

* At N=25 balanced accuracy = 50% (coin flip). Model predicts calm on every covered day.

Comparison to Naive Baselines (New)

Central finding: a coverage-matched persistence rule and a 3-feature HAR logistic regression both match the 36-feature RF. No gap is statistically significant.

N	Coverage	Persistence	HAR-LR (3 feat)	RF Selective	RF minus Persist
5	76.3%	91.6%	91.6%	91.9%	+0.3 pp
10	60.1%	88.1%	91.1%	90.0%	+1.9 pp
15	51.3%	87.9%	90.2%	90.0%	+2.1 pp

20	31.7%	90.5%	94.2%	92.7%	+2.3 pp
25	26.9%	83.8%	92.0%	90.3%	+6.5 pp

At $N \geq 10$, RF and persistence make identical predictions on every jointly covered day (McNemar $b=c=0$). The accuracy gap is entirely explained by which days each rule covers, not forecasting skill. Bootstrap 95% CI at $N=10$: RF vs. persistence $[-1.11, +4.17]$; HAR vs. RF $[-1.08, +4.18]$.

Persistence Quantification (New)

N	P(High High today)	P(High Calm today)	Persistence Ratio	Same-Regime % of RF Correct
5	84.2%	8.6%	9.8×	99.0%
10	79.5%	11.2%	7.1×	99.6%
15	75.9%	13.2%	5.8×	99.8%
20	73.1%	14.8%	4.9×	100.0%
25	70.0%	16.5%	4.2×	100.0%

At $N \geq 20$, 100% of RF correct predictions are same-regime calls. The model adds no directional forecasting beyond VIX autocorrelation.

Regime Transition Analysis (New)

The model achieves 0% accuracy on calm→high spike days at every horizon. The confidence filter covers 87% of calm-persisting days but systematically abstains or makes wrong predictions on spike days. Cost-weighting up to 20× does not fix this — the failure is informational, not objective-function-driven.

Cross-Task Replications (New)

Task	Persistence Ratio (N=5)	Transition Acc	RF minus Persistence
VIX regime	9.8×	0%	+0.3 pp (not significant)
S&P 500 trend regime	23.8×	0%	−4.0 pp (persistence wins)
Yield curve inversion	75.5×	0%	−53.0 pp (persistence wins)
Synthetic (oracle injected)	4.4×	54.9%	+8.7 pp (genuine skill detected)

Severity of the illusion scales with the persistence ratio. The synthetic positive control confirms the diagnostic has power: when a real predictive signal is injected, the protocol correctly identifies it.

Feature Importance (SHAP, New)

VIX technical features (moving averages, Bollinger bands) account for 71.9% of model SHAP mass at $N=5$ and 64.7% at $N=10$. S&P; 500 features are a secondary share (~19–21%). Macro and cross-asset features account for under 10% and are negligible. The model overwhelmingly learns VIX autocorrelation.

Current Status

Paper is 53 pages double-spaced. All results are complete and consistent across three replication tasks. Goal is conference submission (ICAIF, QuantFin, or similar ML-finance venue). Next step: identify target venue and cut to submission length.